TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.8 ± 0.98
Correlation Stability	86.5 ± 1.02
Distribution Consistency	87.1 ± 1.04
Entropy Preservation	81.0 ± 0.77
Mean Z Score Drift	0.3 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	18.5 ± 2.54
Data Density	100.0 ± 0.00
Coverage Score	40.5 ± 3.23
Feature Accuracy	70.4 ± 1.62
Novelty Score	0.5 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	4.651	+0.805	20.9%	Warn
Glucose	120.895	126.769	+5.874	4.9%	OK
BloodPressure	69.105	72.446	+3.341	4.8%	OK
SkinThickness	20.536	23.167	+2.630	12.8%	Warn
Insulin	79.799	87.581	+7.781	9.8%	OK
BMI	31.993	34.465	+2.472	7.7%	OK
DiabetesPedigreeFunction	0.472	0.542	+0.070	14.8%	Warn
Age	33.241	35.516	+2.275	6.8%	OK
Outcome	0.349	1.000	+0.651	186.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.8 ± 0.98
Correlation Stability (*2)	86.5 ± 1.02
Distribution Consistency (*3)	87.1 ± 1.04
Entropy Preservation (*4)	81.0 ± 0.77
Mean Z Score Drift (*5)	0.3 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	18.5 ± 2.54
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	40.5 ± 3.23
Feature Accuracy (*10)	70.4 ± 1.62

Novelty Score (*11)	0.5 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7480519480519481	N/A	0.735064935064935	-0.012987012987012991	
F1	0.6192028104436911	N/A	0.65078987093453	0.03158706049083877	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	N/A	0.7285714285714286	-0.00064935064935064	
F1	0.592152063889281	N/A	0.6311559649830528	0.03900390109377191	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.711038961038961	N/A	0.6883116883116883	-0.022727272727272717	
F1	0.5730341956579464	N/A	0.5994081547041372	0.02637395904619074	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	N/A	0.7428571428571429	-0.020779220779220786	*
F1	0.6399852354635025	N/A	0.652152672324618	0.0121674368611156	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	N/A	0.7181818181818181	-0.03831168831168834	*
F1	0.5733680734057942	N/A	0.6541990765359654	0.08083100313017118	*

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	N/A	0.7571428571428571	0.00779220779220795	
F1	0.6076604413342895	N/A	0.678852254972219	0.07119181363792941	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7597402597402597	N/A	0.75	-0.009740259740259749	
F1	0.6207698060879048	N/A	0.6680893484701913	0.04731954238228625	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	N/A	0.72987012987013	-0.042207792207792215	*
F1	0.6262914445612621	N/A	0.6523811972340006	0.026089752672738553	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.672077922077922	N/A	0.6662337662337662	-0.005844155844155863	
F1	0.5135387174755263	N/A	0.5474036284829072	0.033864911007380996	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	N/A	0.6870129870129871	-0.06753246753246754	*
F1	0.5998259463389416	N/A	0.6408484609281067	0.041022514589164914	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	112.0	64.0	34.0	0.0	33.59743707261777	0.542556613562875	21.0	1.0	1.0
2	7.0	143.0	87.0	0.0	0.0	31.27531807957682	0.486461516326472	47.0	1.0	1.0
3	11.0	140.0	82.0	42.0	323.0	38.87783099281884	0.524599327146421	58.0	1.0	1.0
4	3.0	142.0	68.0	26.0	128.0	32.02707285333422	0.435282796835000 76	36.0	1.0	1.0
5	8.0	145.0	79.0	30.0	127.0	33.20020611859969	0.566795513512328 3	50.0	1.0	1.0
6	8.0	142.0	77.0	41.0	338.0	35.01199886945648	0.456579623577338 06	47.0	1.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	121.0	67.0	30.0	154.0	35.01010527328147 5	0.648700706630575 2	31.0	1.0	2.0
2	9.0	117.0	0.0	0.0	0.0	5.373611863272521	0.267696598370643 6	28.0	1.0	2.0
3	0.0	125.0	72.0	0.0	0.0	33.90431802093256 4	0.261641436608182 5	53.0	1.0	2.0
4	8.0	173.0	68.0	39.0	291.0	39.26798836431936 5	0.440628858972646	39.0	1.0	2.0
5	1.0	126.0	48.0	42.0	183.0	37.60262177876291	0.753442361001631	25.0	1.0	2.0
6	11.0	104.0	72.0	0.0	0.0	30.68145702565908 6	0.396617145258792 1	35.0	1.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	2.0	124.0	60.0	0.0	0.0	30.8519474149700 8	0.30032129892562 11	42.0	1.0	3.0
2	2.0	137.0	58.0	0.0	0.0	26.4670389236739 86	0.43878346948919 933	22.0	1.0	3.0
3	3.0	138.0	70.0	0.0	0.0	32.7893845782651 5	0.98083610113897 76	23.0	1.0	3.0
4	9.0	125.0	70.0	25.0	115.0	31.0757690011030 5	0.21534223089283 444	41.0	1.0	3.0
5	2.0	118.0	80.0	0.0	0.0	40.5914898010945 1	0.62390390163275 98	21.0	1.0	3.0
6	2.0	131.0	86.0	39.0	100.0	35.8503496407476 45	0.72307971534429	36.0	1.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	114.0	76.0	32.0	0.0	36.2869929110770 3	0.73286573251278 72	37.0	1.0	4.0
2	9.0	102.0	92.0	36.0	0.0	44.4860779427645 74	0.70157039767818 33	41.0	1.0	4.0
3	3.0	165.0	70.0	38.0	219.0	36.1705639270266 56	0.36929632513347 75	29.0	1.0	4.0
4	4.0	102.0	73.0	0.0	0.0	36.922805062027	0.34640693124474 164	40.0	1.0	4.0
5	7.0	132.0	72.0	0.0	0.0	33.0181691056657 7	0.28039888129858 78	39.0	1.0	4.0
6	7.0	169.0	74.0	32.0	490.0	30.1855491601325 63	0.47612519671813 96	55.0	1.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	145.0	79.0	7.0	0.0	35.8814447527925	0.3956186257576436	68.0	1.0	5.0
2	1.0	129.0	99.0	47.0	144.0	49.00899181927188	0.7014450915079007	29.0	1.0	5.0
3	4.0	135.0	83.0	39.0	158.0	33.10522395388794	0.5513481125020735	27.0	1.0	5.0
4	2.0	114.0	73.0	27.0	0.0	30.583678034819375	0.2107929409560228	27.0	1.0	5.0
5	0.0	117.0	82.0	47.0	226.0	44.55734843682979	0.6708373174137706	34.0	1.0	5.0
6	2.0	122.0	93.0	0.0	0.0	42.75059440061146	0.4634498868579099	29.0	1.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	8.0	99.0	94.0	34.0	0.0	40.171066978881	0.45728032742562597	49.0	1.0	6.0
2	7.0	108.0	80.0	31.0	0.0	35.863346646916064	1.1159306873686852	43.0	1.0	6.0
3	3.0	112.0	74.0	0.0	0.0	28.227059825137243	0.3345610523900302	48.0	1.0	6.0
4	4.0	139.0	80.0	36.0	161.0	32.931475753655562	0.3586125262348244	26.0	1.0	6.0
5	4.0	146.0	78.0	0.0	0.0	38.171256360127813	0.5208556341311742	66.0	1.0	6.0
6	2.0	146.0	82.0	41.0	174.0	43.296761560472675	0.5378784429962995	45.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	129.0	83.0	0.0	0.0	37.97263978373522	0.473849755650675	43.0	1.0	7.0
2	2.0	122.0	68.0	28.0	205.0	33.27344480348626	0.862705048008298	30.0	1.0	7.0
3	2.0	147.0	70.0	37.0	356.0	28.840260502318599	0.33778424313549744	29.0	1.0	7.0
4	8.0	144.0	88.0	33.0	164.0	30.382714095396558	0.753233012308819 2	52.0	1.0	7.0
5	9.0	90.0	85.0	32.0	0.0	35.420641740094355	0.788679040512465 7	55.0	1.0	7.0
6	7.0	121.0	82.0	0.0	0.0	29.823901931784874	0.249083540117926 74	26.0	1.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	146.0	78.0	0.0	0.0	35.159608569581714	0.3930651256441051	60.0	1.0	8.0
2	4.0	133.0	72.0	0.0	0.0	24.396508787995 34	0.2676944629072727	59.0	1.0	8.0
3	1.0	139.0	80.0	43.0	180.0	45.331024423927 63	0.5791497454029765	41.0	1.0	8.0
4	8.0	118.0	75.0	35.0	0.0	30.1293989860443	0.2586746421811069	31.0	1.0	8.0
5	7.0	51.0	82.0	32.0	0.0	37.483014712115086	0.6221698283437497	47.0	1.0	8.0
6	5.0	146.0	80.0	1.0	0.0	29.770139755663436	0.1831326553576052 8	51.0	1.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	12.0	151.0	87.0	33.0	25.0	27.727368055376672	0.71681672385894 5	43.0	1.0	9.0
2	3.0	135.0	70.0	0.0	0.0	30.803132386490205	1.07156727276249 2	22.0	1.0	9.0
3	5.0	102.0	81.0	0.0	0.0	30.898364398658387	0.18127760655728 703	36.0	1.0	9.0
4	4.0	159.0	81.0	40.0	211.0	38.866035521812044	0.39075007864299 377	29.0	1.0	9.0
5	2.0	130.0	64.0	28.0	117.0	28.664488866838446	0.24863405183763 912	27.0	1.0	9.0
6	4.0	146.0	81.0	19.0	0.0	32.488456467255624	0.25901054810830 687	67.0	1.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	7.0	107.0	72.0	0.0	0.0	30.048176868691266	0.60307998328063 73	31.0	1.0	10.0
2	7.0	93.0	80.0	29.0	0.0	35.26469389855039	0.59211689621139 56	53.0	1.0	10.0
3	1.0	132.0	101.0	27.0	140.0	32.871166223397455	0.24107475985539 43	44.0	1.0	10.0
4	4.0	129.0	0.0	0.0	0.0	27.419068698819655	0.29516966008059 59	24.0	1.0	10.0
5	7.0	125.0	95.0	0.0	0.0	0.7708384606019952	0.23958485280143 632	54.0	1.0	10.0
6	1.0	116.0	78.0	0.0	0.0	39.462881594378565	0.54441623559032 35	21.0	1.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan